Aseem Baji

Applied Scientist

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Links

LinkedIn, Github

Professional Summary

Information Systems PhD candidate studying machine learning (ML) pipeline design, data preprocessing, hyperparameter optimization, and budget-aware model evaluation. Lead student researcher on a large-scale machine learning experimentation project evaluating approximately 3.6M candidate pipelines across 30 supervised prediction tasks. Strong background in R, Python, SQL, Tableau, causal inference, econometrics, A/B testing, experimental design, statistical analysis, and business analytics. Prior analytics consulting experience applying machine learning and data science to e-commerce, consumer packaged goods (CPG), transportation, and forecasting problems.

TECHNICAL SKILLS

- Python
- Predictive Modeling
- Recommender Systems
- Causal Inference
- Experimental Design
- Power BI
- High-performance Computing (HPC)
- R
- Supervised & Unsupervised Learning
- Model Evaluation
- Econometrics
- Fairness-aware Machine Learning
- SQL Server
- Simple Linux Utility for Resource Management (SLURM) job arrays
- SQL
- Forecasting
- Hyperparameter Optimization
- A/B testing
- Tableau
- Relational Database Design

WORK EXPERIENCE

Fractal Analytics, Mumbai, India Jun 2017 to Aug 2020
Consultant

- Identified approximately $500K in white-space opportunities for a leading CPG manufacturer's Amazon business and led a 3-person analytics team supporting a $22M Amazon portfolio to prioritize high-impact actions.
- Redesigned a cross-team classification workflow, cutting processing time by 25% and improving classification accuracy by 5%, which accelerated downstream analytics and decision-making.
- Developed an SKU-level recommender to surface product-level sales opportunities and guide account prioritization, extending the engagement beyond initial scope and informing merchandising strategies.
- Delivered client-facing predictive models, forecasts, and executive dashboards, translating technical results into commercial recommendations that informed pricing, assortment, and promotion decisions.

University of Maryland (UMD) Center for Advanced Transportation Technology, College Park, MD May 2021 to Aug 2022
Data Scientist

- Designed and implemented R and Tableau analysis pipelines and a parallelized tool to analyze traffic incidents and weather-related crashes across all 52 U.S. states.
- Evaluated feasibility of time-series and regression methods for transportation metrics with limited observations and recommended robust approaches for trend and indicator selection.

- Produced reproducible analyses and visualizations to support transportation stakeholders and inform data-driven safety recommendations.

RESEARCH EXPERIENCE

Doctoral Researcher - Machine-Learning Pipelines, Preprocessing & Fairness, College Park, MD Aug 2022 to Present

University of Maryland

- Led a budget-aware ML study evaluating ~3.6M candidate pipelines across 30 supervised tasks to measure effects of data preprocessing and hyperparameter optimization (HPO).
- Found joint preprocessing + HPO outperformed preprocessing-only in 23/30 contexts and HPO-only in 26/30 contexts.
- Scaled experiments with scikit-learn, XGBoost, Optuna on UMD HPC (up to 1,000 parallel jobs) and produced analysis-ready trial logs; developed R workflows for efficiency curves, validation-test gap analysis, statistical testing, and rank-stability assessments.
- Extending the framework to fairness and LLM pretraining/fine-tuning.

Applied Artificial Intelligence (AI) Researcher - Large Language Model (LLM)-Based Personalized Math Feedback 2026 to Present

University of Maryland/Akanksha Foundation

- Working with Akanksha foundation, a non-profit which runs schools in underprivileged communities to explore a research project on leveraging LLMs to create customized feedback at scale.
- Leading development of personalized practice from item-level math responses for Grades 5–8 using open-source LLMs and teacher-in-the-loop review.
- Preparing a school-level pilot to measure downstream learning outcomes post delivery of LLM generated feedback.

TEACHING EXPERIENCE

Instructor - BMGT 402: Database Systems, College Park, MD Jan 2024 to Jan 2025

- Sole primary instructor for two undergraduate sections serving 70 students; designed and delivered SQL labs, assignments, office hours, a semester-long database-design project, and final assessment.
- Taught SQL, SQL Server, entity-relationship modeling, relational design, normalization, query processing, data warehousing, data mining, and business information systems.

EDUCATION

PhD, Information Systems Aug 2022 to May 2027

University of Maryland, Robert H. Smith School of Business, College Park, MD

Dissertation: Making Experimental Effort Visible in Machine-Learning Pipelines: Data Preprocessing, Fairness, and LLM Training | Advisor: Dr. Jessica Clark

MS, Business Analytics Sept 2020 to Dec 2021

University of Maryland, Robert H. Smith School of Business, College Park, MD

Terrapin Fellow

BE (Hons.), Manufacturing Aug 2013 to Jul 2017

BITS Pilani, Hyderabad Campus, Hyderabad, India

Accomplishments

Peer-reviewed conference presentations (selected):

- Symposium on Statistical Challenges in Electronic Commerce Research (SCECR), 2026 — Tokyo, Japan
- Symposium on Statistical Challenges in Electronic Commerce Research (SCECR), 2024 — Lisbon, Portugal
- Symposium on Statistical Challenges in Electronic Commerce Research (SCECR), 2023 — Bogota, Colombia
- Workshop on Information Technologies and Systems (WITS), 2024 — Bangkok, Thailand

Awards

Ritu Agarwal Endowed Distinguished Ph.D. Fellowship in Information Systems, University of Maryland Oct 2026

Jacob K. Goldhaber Travel Grant, University of Maryland May 2026

International Conference Student Support Award, University of Maryland Nov 2024